Exercise 2 SQL Fundamentals.

Aggregate Functions & Grouping

Question 1

Count() function

Write SQL query to find the total number of employees in the company.

~~Select count(first_name, last_name)~~
~~from employees;~~

```
Select count (id)
from employees;
```

id
10

Question 2. Sum( ) function. Write a SQL query to find the total salary paid to all employees in the IT department.

```
Select sum (salary), department
from employees
where department = 'IT';
```

department	Salary
IT	270'000

Question 3 AVG( ) function: Write SQL query to calculate the average salary of employees in the HR department

```
Select avg(salary), department
from employees
where department = 'HR';
```

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department	salary
HR	49500

Question 4 MIN() and MAX() functions : Write SQL to find the highest and lowest salary in the company

```
Select max(salary),
       min (salary)
from employees;
```

Salary
62 000
48 000

Question 5 GROUP BY ~~and COUNT()~~ statement :
write SQL query to group employees by department and display the total salary paid in each department

```
Select sum(salary) as total_salary, department
~~select sum(salary), department~~
from employees
group by department;
```

~~department~~	~~total salary~~

department	total_salary
IT	220 000
HR	99 000
Finance	119 000
Marketing	105000

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Question 6 Group by and count()

Write SQL query to ~~group~~ count how many employees work in each city

```
select count(id) as number_of_employees, city
~~Select count(id), city~~
from employees
group by city;
```

Number_of_employees ~~id~~	City
2	Newyork
3	Chicago
2	Los Angeles
2	San fransisco
1	Houston

Question 7 Groupby & order by : write a sql query to ~~list~~ group employees by department, calculate average salary in each department, and order the results in descending order of average salary

```
select avg(salary) as avg_salary, department
from employees
group by department
order by department desc;
```

department	avg_salary
IT	55000
HR	49500
Finance	59 500
Marketing	52500

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Question 8 Having clause

Write a sql query to find departments where the total salary paid exceeds 100,000.

```
Select sum(salary) as total_salary, department
from employees
groupby department
having sum(salary) > 100 000;
```

department	total_salary
IT	220000
Finance	119 000
Marketing	105 000

9. ~~combin~~ Question 9 Combining groupby Having & or

write ~~or~~ sql query to list cities where more than one employee works, ordered by the number of empl descending

```
Select count(id), ~~city~~ as number_of_employees,
from employees
groupby city
having count(id) > 1
orderby number_of_employees desc;
```

Number of	

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number_of_employees	city
3	Chicago
2	New York
2	Los Angeles
2	San Francisco

Question 10 : combining aggregate functions

write sql query to find the department with the highest average salary.

```
select avg(salary) as avg_salary, department
from employees
group by department
orderby avg_salary desc
limit 1 ;
```

avg_salary	department
Finance	59500

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